

that this attribute is creating a relationship with the primary key of an other table and is the relation link between these tables or entities.

If two horizontal lines are used at the beginning of relation lines between entities, these lines indicate one relationship to the other table. If the other ending of the relation line is also two horizontal line, the relationship between these two tables are one to one. If the other ending of the relation line is crow's foot with a line, then it is one to many relationship.

Part II: Entities and their Attributes Explanation

Airport entity: is the table where data about a specific airport is stored. And its attributes are the airport-id (is a unique number that identifies each airport and it cannot be null or repeated. Second, airport-name is the formal name of the specific airport. It can be text or varchar when creating the database. Additionally, this attribute also cannot be null. The other attributes are country, in which country the airport is located, in which state, if there are any state in that country, and in which city of that country the airport is located. All of these three attributes can be text or varchar. The last two attributes of this entity are created-at and updated-at. Created-at stores the exact time that this airport database was created for the first time. And updated-at stores the time that any changes have been made to this databases of airport entity.

Airline entity: stores information about the airlines that this specific airport has, like to which countries or states the airport has flights. Its first attribute is airline-id, which is primary key (numeric value that should be unique for each airline and not null.) The airline-code is also unique for each airline and it can be text. Then name is the name of airline that can be text or varchar, such as Emirates. The country attribute stores information of where the airline is registered. The last two attributes of this entity are created-at and updated-at. Created-at stores the exact time that this airport database was created for the first time. And updated-at stores the time that any changes have been made to these databases of airport entity.

Flight entity: it stores key information about the flights that are made by an airline. Attributes flight-id is a primary key, the airline-id is a foreign key that creates relationship between flight and airline entities. This relationship is one to many for airline to flights, because an airline is unique, but it can have many different flights. The attributes of departure-airport-id, arrival-airport-id are foreign keys that are one to many, because one flight has only one departure or arrival-id, but and airport can be the departure of many deferent flights over time. They create relationship with airport entity from flight entity.

And that applies to other entities in the database as well.

Part III: Reasons of explaining why specific kind of them.

Relationship from airline to flight is one to many because. Every flight is controlled only by one airline company; however an airline does not control one flight, it can have thousands of flights depending on the airline.

Relationship from airport to flights both departure and arrival are one to many because airport can have many both departing and arriving flights, each flight can depart from one place and arrive in only one place.

Flights to booking: an aircraft has many seats for each time that it flights so many booking can be done to one flight. So the relationship is one from flight to many booking.

Passenger to booking: one passenger can book many different flights over time, however a single booking transaction is only for one passenger. No two passenger can have the same booking transaction.

Booking to Book-change entities relationship is one to many because a booking can be changed many times, but still it belongs to that first booking that was the first.

Booking to Boarding Pass: the relationship is one to one because each booking receives only one boarding pass which is also applicable only for that specific booking not any other booking.

Booking-baggage: these entities relationship is one to many because, for example, 3 baggage can be listed in one booking ticket. However a baggage cannot be related to more than one booking.

Baggage- baggage check: so every baggage in the airport is Each checked item of luggage goes through the terminal screening lines precisely once per trip, creating a single verification record tracking the item's safety clearance.

Passenger to security check: A passenger must clear security. Over an extended timeframe, a frequent flyer will generate a long historical record of distinct security checkpoint clear entries, though each entry belongs to exactly one individual.